Lecture 2

The Fundamentals of Electronics: A Review

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Topics Covered (Chapter 2)

- Gain, Attenuation, and Decibels
- Tuned Circuits
- Filters
- Fourier Theory

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2-1: Gain, Attenuation, and Decibels

- Most circuits in electronic communication are used to manipulate signals to produce a desired result.

- All signal processing circuits involve:
  - Gain
  - Attenuation

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2-1: Gain, Attenuation, and Decibels

Gain

- **Gain** means amplification. It is the ratio of a circuit's output to its input.

$$A_V = \frac{\text{output}}{\text{input}} = \frac{V_{out}}{V_{in}}$$

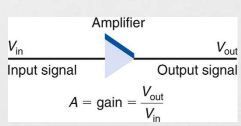

Figure 2-1: An amplifier has gain.

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2-1: Gain, Attenuation, and Decibels

- Most amplifiers are also power amplifiers, so the same procedure can be used to calculate power gain A_p where P_{in} is the power input and P_{out} is the power output.

$$\text{Power gain } (A_p) = P_{out} / P_{in}$$

- **Example:**

The power output of an amplifier is 6 watts (W). The power gain is 80.

What is the input power?

$$A_p = P_{out} / P_{in} \quad \text{therefore} \quad P_{in} = P_{out} / A_p$$
$$P_{in} = 6 / 80 = 0.075 \text{ W} = 75 \text{ mW}$$

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2-1: Gain, Attenuation, and Decibels

- An amplifier is **cascaded** when two or more stages are connected together.
- The overall gain is the product of the individual circuit gains.

- **Example:**

Three cascaded amplifiers have power gains of 5, 2, and 17.
The input power is 40 mW. What is the output power?

$$A_p = A_1 \times A_2 \times A_3 = 5 \times 2 \times 17 = 170$$
$$A_p = P_{out} / P_{in} \quad \text{therefore} \quad P_{out} = A_p P_{in}$$
$$P_{out} = 170\ (40 \times 10^{-3}) = 6.8\text{W}$$

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2-1: Gain, Attenuation, and Decibels

Attenuation

- **Attenuation** refers to a loss introduced by a circuit or component. If the output signal is lower in amplitude than the input, the circuit has loss or attenuation.
- The letter A is used to represent attenuation
- Attenuation A = output/input = V_{out}/V_{in}
- Circuits that introduce attenuation have a gain that is less than 1.
- With cascaded circuits, the total attenuation is the product of the individual attenuations.

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2-1: Gain, Attenuation, and Decibels

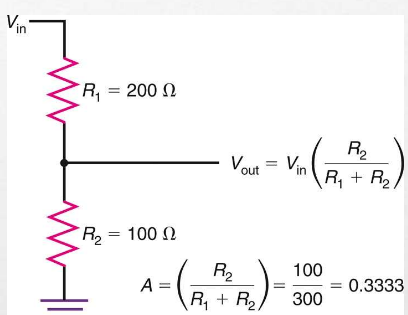

Figure 2-3: A voltage divider introduces attenuation.

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2-1: Gain, Attenuation, and Decibels

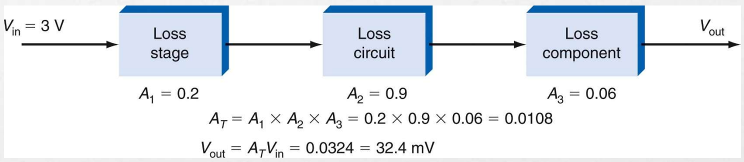

Figure 2-4: Total attenuation is the product of individual attenuations of each cascaded circuit.

2-1: Gain, Attenuation, and Decibels

Decibels

- The **decibel (dB)** is a unit of measure used to express the gain or loss of a circuit.
  - The decibel was originally created to express hearing response.
  - A decibel is one-tenth of a bel.
- When gain and attenuation are both converted into decibels, the overall gain or attenuation of a circuit can be computed by adding individual gains or attenuations, expressed in decibels.

2-1: Gain, Attenuation, and Decibels

Decibels: Decibel Calculations

- Voltage Gain or Attenuation

  $$dB = 20 \log V_{out} / V_{in}$$

- Current Gain or Attenuation

  $$dB = 20 \log I_{out} / I_{in}$$

- Power Gain or Attenuation

  $$dB = 10 \log P_{out} / P_{in}$$

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2-1: Gain, Attenuation, and Decibels

Decibels: Decibel Calculations

- **Example**:

  An amplifier has an input of 3 mV and an output of 5 V. What is the gain in decibels?

  dB = 20 log 5/0.003
  = 20 log 1666.67
  = 20 (3.22)
  = 64.4

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2-1: Gain, Attenuation, and Decibels

Decibels: Decibel Calculations

- **Example:**

  A filter has a power input of 50 mW and an output of 2 mW. What is the gain or attenuation?

  $$\begin{aligned} dB &= 10 \log (2/50) \\ &= 10 \log (0.04) \\ &= 10 (-1.398) \\ &= -13.98 \end{aligned}$$

  - If the decibel figure is positive, that denotes a gain.

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2-1: Gain, Attenuation, and Decibels

Decibels: Antilogs

- The **antilog** is the number obtained when the base is raised to the logarithm which is the exponent.
- Antilogs are used to calculate input or output voltage or power, given the decibel gain or attenuation and the output or input.
- The antilog is the base 10 raised to the dB/10 power.
- The antilog is readily calculated on a scientific calculator.

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2-1: Gain, Attenuation, and Decibels

Decibels: dBm and dBc

- When a decibel value is computed by comparing a power value to 1 mW, the result is a value called the **dBm**. This is a useful reference value.
- The value **dBc** is a decibel gain attenuation figure where the reference is the carrier.

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2-2: Tuned Circuits

- Virtually all communications equipment contains **tuned circuits** made up of inductors and capacitors that resonate at specific frequencies.

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2-2: Tuned Circuits

Tuned Circuits and Resonance

- A **tuned circuit** is made up of inductance and capacitance and resonates at a specific frequency, the **resonant frequency**.
- The terms **tuned circuit** and **resonant circuit** are used interchangeably.
- Tuned circuits are frequency-selective and respond best at their resonant frequency.

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2-2: Tuned Circuits

Tuned Circuits and Resonance: Series Resonant Circuits

- A **series resonant circuit** is made up of inductance, capacitance and resistance connected in series.
- Series resonant circuits are often referred to as *LCR* or *RLC* circuits.
- Resonance occurs when inductive and capacitive reactances are equal.
- Resonant frequency (f_r) is inversely proportional to inductance and capacitance.

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

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2-2: Tuned Circuits

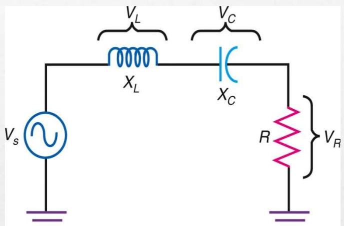

Figure 2-13: Series *RLC* circuit.

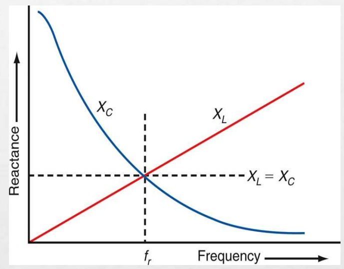

Figure 2-14 Variation of reactance with frequency.

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2-2: Tuned Circuits

Tuned Circuits and Resonance: Series Resonant Circuits

- **Example:**

  What is the resonant frequency of a 2.7-pF capacitor and a 33-nH inductor?

$$f_r = 1/2\pi\sqrt{LC}$$

$$= 1/6.28\sqrt{33 \times 10^{-9} \times 2.7 \times 10^{-12}}$$

$$= 5.33 \times 10^{8} \text{ Hz or } 533 \text{ MHz}$$

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2-2: Tuned Circuits

Tuned Circuits and Resonance: Series Resonant Circuits

- The **bandwidth (BW)** of a series resonant circuit is the narrow frequency range over which the current is highest.
- **Half-power points** are the current levels at which the frequency response is 70.7% of the peak value of resonance.
- The **quality (*Q*)** of a series resonant circuit is the ratio of the inductive reactance to the total circuit resistance.
- **Selectivity** is how a circuit responds to varying frequencies.
- The bandwidth of a circuit is inversely proportional to *Q*.

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2-2: Tuned Circuits

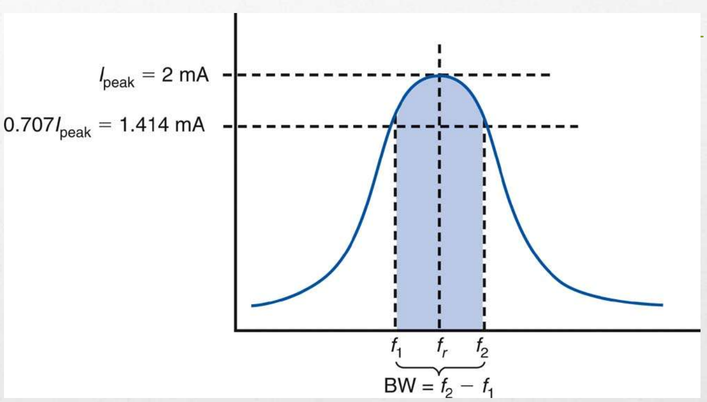

Figure 2-16: Bandwidth of a series resonant circuit.

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2-2: Tuned Circuits

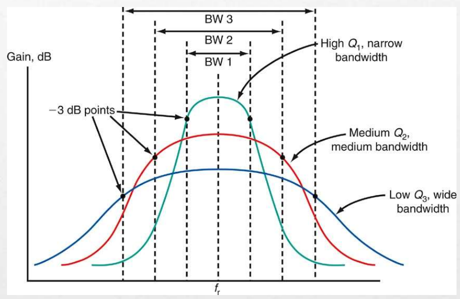

Figure 2-17: The effect of *Q* on bandwidth and selectivity in a resonant circuit.

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2-2: Tuned Circuits

Tuned Circuits and Resonance: Parallel Resonant Circuits

- A **parallel resonant circuit** is formed when the inductor and capacitor of a tuned circuit are connected in parallel with the applied voltage.
- A parallel resonant circuit is often referred to as a *LCR* or *RLC* circuit.
- Resonance occurs when inductive and capacitive reactances are equal.
- The resonant frequency (f_r) is inversely proportional to inductance and capacitance.

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2-2: Tuned Circuits

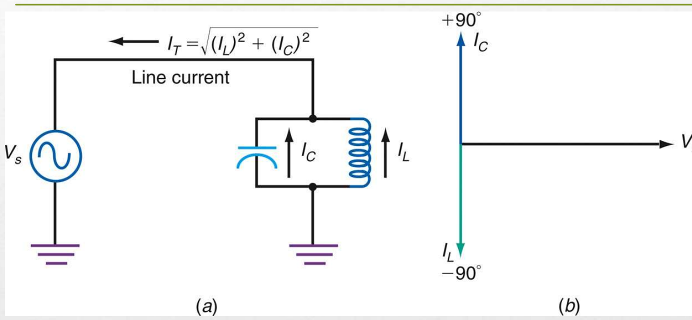

Figure 2-19: Parallel resonant circuit currents. (*a*) Parallel resonant circuit. (*b*) Current relationships in parallel resonant circuit.

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2-2: Tuned Circuits

Tuned Circuits and Resonance: Parallel Resonant Circuits

- At resonance, a parallel tuned circuit appears to
  - have infinite resistance
  - draw no current from the source
  - have infinite impedance
  - act as an open circuit.
- However, there is a high circulating current between the inductor and capacitor, storing and transferring energy between them.

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2-2: Tuned Circuits

Tuned Circuits and Resonance: Parallel Resonant Circuit

- Because such a circuit acts as a kind of storage vessel for electric energy, it is often referred to as a **tank circuit** and the circulating current is referred to as the **tank current**.

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2-3: Filters

- A **filter** is a frequency-selective circuit.
- Filters pass certain frequencies and reject others.
- **Passive filters** are created using components such as: resistors, capacitors, and inductors that do not amplify.
- **Active filters** use amplifying devices such as transistors and operational amplifiers.

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2-3: Filters

- There are five basic kinds of filter circuits:
  - **Low-pass filters** only pass frequencies below a critical (cutoff) frequency.
  - **High-pass filters** only pass frequencies above the cutoff frequency.
  - **Bandpass filters** pass frequencies over a narrow range between lower and upper cutoff frequencies.
  - **Band-reject filters** reject or stop frequencies over a narrow range between lower and upper cutoff frequencies.
  - **All-pass filters** pass all frequencies over a desired range but have a predictable phase shift characteristic.

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2-3: Filters

LC Filters

- *LC* filters use combinations of inductors and capacitors to achieve a desired frequency response.
- They are typically used with radio frequency (RF) applications.

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2-3: Filters

LC Filters

- **Passband** is the frequency range over which the filter passes signals.
- **Stop band** is the range of frequencies outside the passband; that is, the range of frequencies that is greatly attenuated by the filter.
- **Attenuation** is the amount by which undesired frequencies in the stop band are reduced.

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2-3: Filters

LC Filters

- **Insertion loss** is the loss the filter introduces to the signals in the passband.
- **Impedance** is the resistive value of the load and source terminations of the filter.
- **Ripple** is a term used to describe the amplitude variation with frequency in the passband.
- **Shape factor** is the ratio of the stop bandwidth to the pass bandwidth of a bandpass filter.

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2-3: Filters

LC Filters

- A **pole** is a frequency at which there is a high impedance in the circuit.
- **Zero** is a term used to refer to a frequency at which there is zero impedance in the circuit.
- **Envelope delay** or **time delay** is the time it takes for a specific point on an input waveform to pass through the filter.
- **Roll-off** or **attenuation rate** is the rate of change of amplitude with frequency in a filter.

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2-3: Filters

Types of Filters

- The most widely used *LC* filters are named after the people who discovered them and developed the analysis and design method for each.
  - Butterworth: The Butterworth filter effect has maximum flatness in response in the passband and a uniform attenuation with frequency.
  - Chebyshev: Has extremely good selectivity, and attenuation just outside the passband is very high, but has ripple in the passband.
  - Cauer (Elliptical): Produces greater attenuation out of the passband, but with higher ripple within and outside of the passband.
  - Bessel (Thomson): Provides the desired frequency response (i.e., low-pass, bandpass, etc.) but has a constant time delay in the passband.

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2-3: Filters

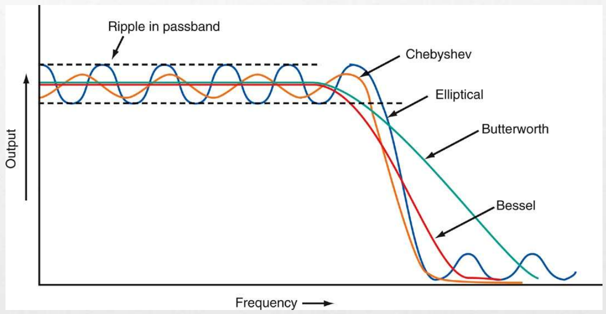

Figure 2-34: Butterworth, elliptical, Bessel, and Chebyshev response curves.

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2-3: Filters

Active Filters

- **Active filters** are frequency-selective circuits that incorporate *RC* networks and amplifiers with feedback to produce low-pass, high-pass, bandpass, and bandstop performance. Advantages are: Gain, No inductors, Easy to tune, Isolation, and Easier impedance matching.
- A special form of active filter is the variable-state filter, which can simultaneously provide low-pass, high-pass, and bandpass operation from one circuit.

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2-4: Fourier Theory

- One method used to determine the characteristics and performance of a communication circuit or system, specifically for non-sine wave approach, is **Fourier analysis.**
- The Fourier theory states **that a nonsinusoidal waveform can be broken down into individual harmonically related sine wave or cosine wave components.**
- A **square wave** is one classic example of this phenomenon.

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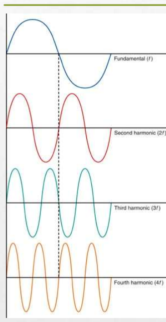
2-4: Fourier Theory



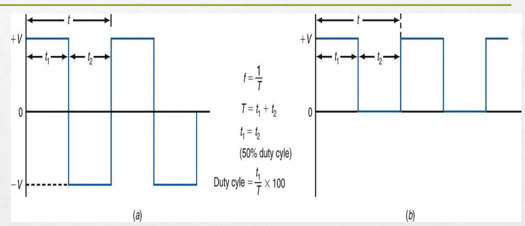


Figure 2-58: A square wave.

Figure 2-57: A sine wave and its harmonics.

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2-4: Fourier Theory

Basic Concepts

- Fourier analysis states that a square wave is made up of a sine wave at the fundamental frequency of the square wave plus an infinite number of odd harmonics.
- Fourier analysis allows us to determine not only sine-wave components in a complex signal but also a signal's bandwidth.

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2-4: Fourier Theory

Time Domain Versus Frequency Domain

- Analysis of variations of voltage, current, or power with respect to time are expressed in the **time domain**.
- A **frequency domain** plots amplitude variations with respect to frequency.
- Fourier theory gives us a new and different way to express and illustrate complex signals, that is, with respect to frequency.

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2-4: Fourier Theory

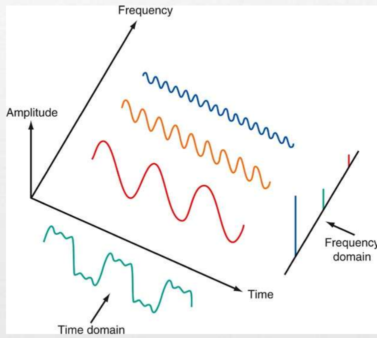

Figure 2-63: The relationship between time and frequency domains.

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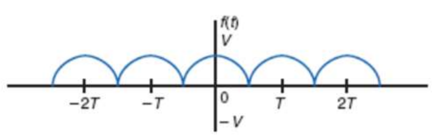
2-4: Fourier Theory



$$f(t) = \frac{2V}{\pi} + \frac{2V}{\pi}\left[\frac{2}{3}\cos 2\pi\left(\frac{1}{T}\right) - \frac{2}{15}\cos 2\pi\left(\frac{2}{T}\right)t + \frac{2}{35}\cos 2\pi\left(\frac{3}{T}\right)t + \ldots\right]$$

(*e*)

Figure 2-61: Common nonsinusoidal waves and their Fourier equations. (*e*) Full cosine wave.

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2-4: Fourier Theory

Time Domain Versus Frequency Domain

- A **spectrum analyzer** is an instrument used to produce a frequency-domain display.
- It is the key test instrument in designing, analyzing, and troubleshooting communication equipment.

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